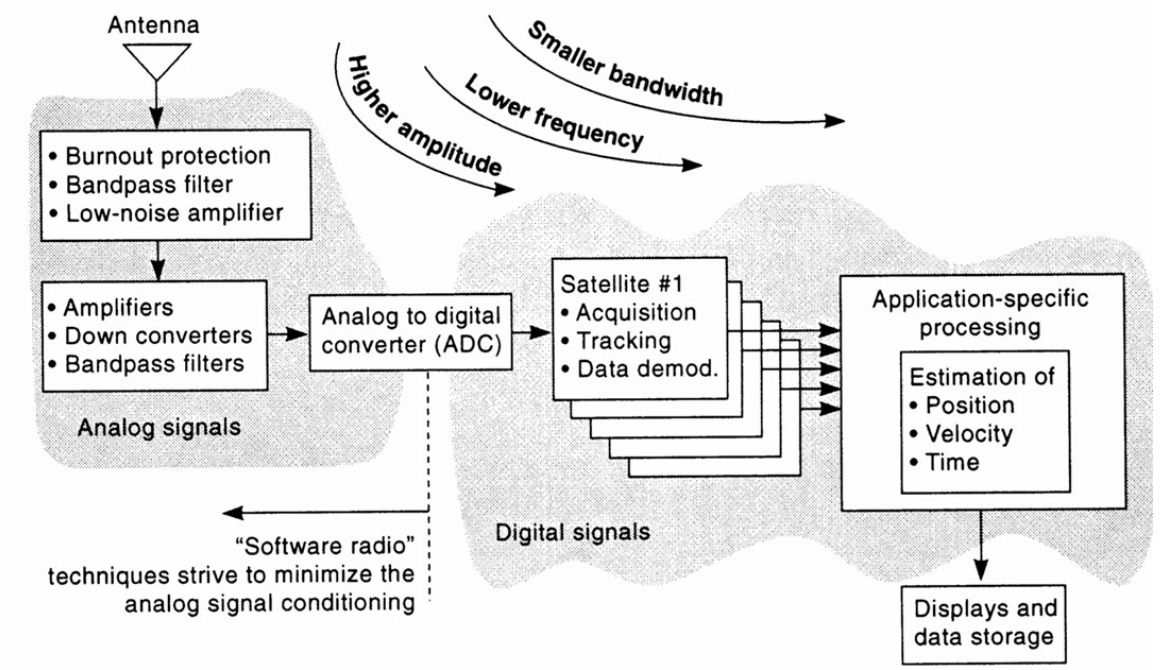
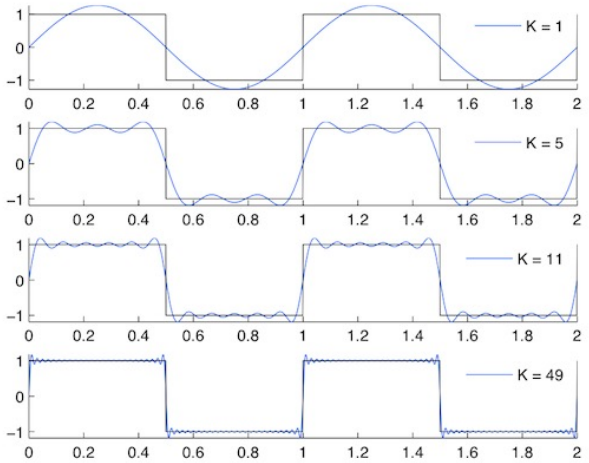
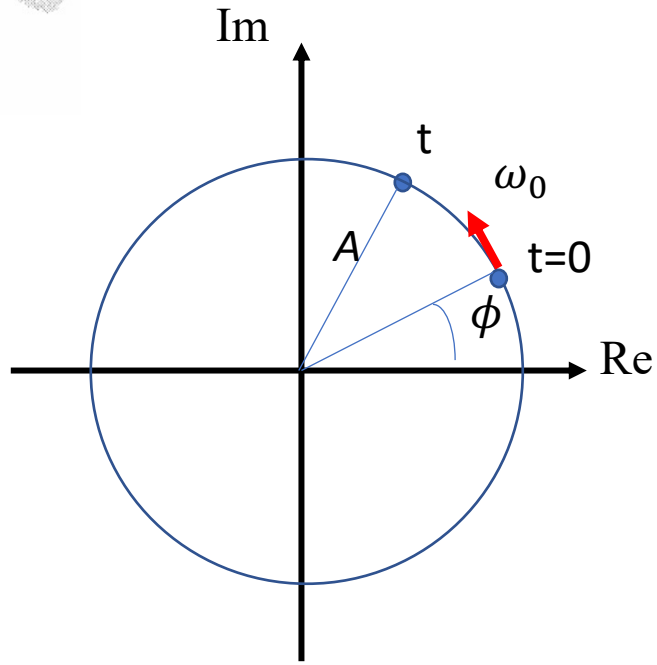
ASEN 5090
Lecture 17
Introduction to Signals & Spectra







Penny Axelrad

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Where are we?

Fundamentals/Basics

Measurements & Position Solution

Where do measurements come from?

Signals, Receivers Acquisition & Tracking

Applications

Week	Date	Topic	Reading	Assignments - Due 11pm*
1	8/24/26	L1. Introduction & GPS Basics	PNT Ch 1, ME 2.1 & 2.2	
	8/26/26	L2. Navigation Measurements & Methods	ME Ch 1	
	8/28/26			HW1 Assigned
2	8/31/26	L3. GPS Signal Structure	ME Ch 2, PNT Ch 2	
	9/2/26	L4. Coordinate Frames	ME Ch 4.1, 4.A	
	9/4/26			HW1 Due, HW2 Assigned
3	9/7/26	Labor Day (no class)		
	9/9/26	L5. Orbits & Visibility	ME Ch 4.3	
	9/11/26			HW3 Assigned
4	9/14/26	L6. Time Systems, Clocks, & Relativity	ME 4.2, Ashby 2003	
	9/16/26	L7. Measurement Model & IGS Data	ME 5.1, 5.2, https://igs.org/, PNT Ch 27.2	
	9/18/26			HW2 Due, HW4 Assigned
5	9/21/26	L8. Ionospheric Effects & Models	ME Ch 5.3.2, PNT Ch 31	
	9/23/26	L9. Tropospheric Effects and Models	ME Ch 5.3.3	
	9/25/26			HW3 Due
6	9/28/26	L10. Multipath	ME Ch 5.4.2+ PNT Ch 22	
	9/30/26	L11. Error Mitigation & Summary	ME 5.6, 5.8	
	10/2/26			HW4 Due, HW5 Assigned
7	10/5/26	L12. Least Squares Solutions	ME Ch 6.A	
	10/7/26	L13. GPS Point Solutions	ME Ch 6.1	
	10/9/26			
8	10/12/26	L14. Improved GPS Solutions		
	10/14/26	L15. Midterm Review & Homework Discussion		
	10/16/26			HW5 Due, HW6 Assigned
9	10/19/26	L16. Precise Positioning Methods	ME Ch 7	
	10/21/26	Midterm Exam (no lecture)		Midterm Release 10/21 8AM
	10/23/26			Midterm Close 10/25 11PM
10	10/26/26	L17. Introduction to Signals & Spectra	ME Ch 8	
	10/28/26	L18. GNSS Spectrum	ME Ch 9	
	10/30/26			
11	11/2/26	L19. Receiver Overview, Antennas, Link Budget	ME Ch 11, PNT Ch 14	HW6 Due, HW7 Assigned
	11/4/26	L20. GNSS Receiver Architecture	ME Ch 10 (parts)	
	11/6/26			
12	11/9/26	L21. GPS Acquisition	ME Ch 11.2	
	11/11/26	L22. GPS Acquistion (continued)		
	11/13/26			HW7 Due, HW8 Assigned
13	11/16/26	L23. GPS Signal Tracking Basics	ME Ch 12, PNT Ch 15	
	11/18/26	L24. Advanced Signal Tracking (Xu)		
	11/20/26			
14	11/23/26	FALL BREAK		
	11/25/26			
	11/27/26			
15	11/30/26	L25. Space Applications of GNSS	PNT Ch 62-63	
	12/2/26	L26. Aircraft and UAV Applications	PNT Ch 60-61	HW8 Due
	12/4/26	L27. Final Exam Review (Monday Class Schedule)		
16	12/7/26	Final Exam available starting Friday, 12/4		Final Release 12/4 6PM
	12/9/26	Final Exam due Tues, 12/8 (24hr time limit)		Final Close 12/8 11PM
	12/11/26			

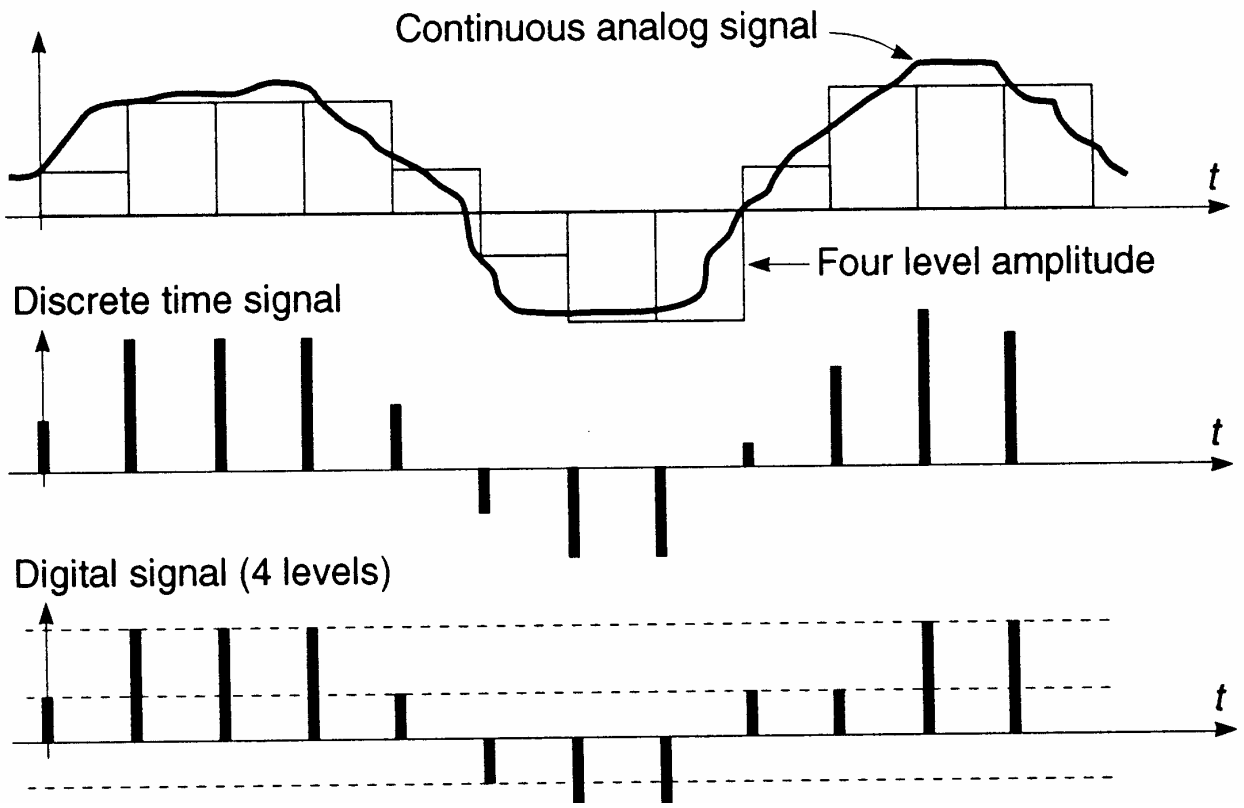
Outline

- Signal Basics
  - Analog, discrete, digital
  - Sinusoids, unit steps, unit pulse, unit impulse
  - Energy and Power
  - A/D Conversion
- Fourier series of periodic functions
  - Sine/Cosine
  - Square wave
  - Simple rectangle pulse train
- Fourier transforms
  - Power spectrum and autocorrelation
- Spectrum of a PRN code

What is a signal?
A function of one or more variables, which represents a physical phenomenon.

Analog, Discrete, Digital Signals

GLOBAL POSITIONING SYSTEM



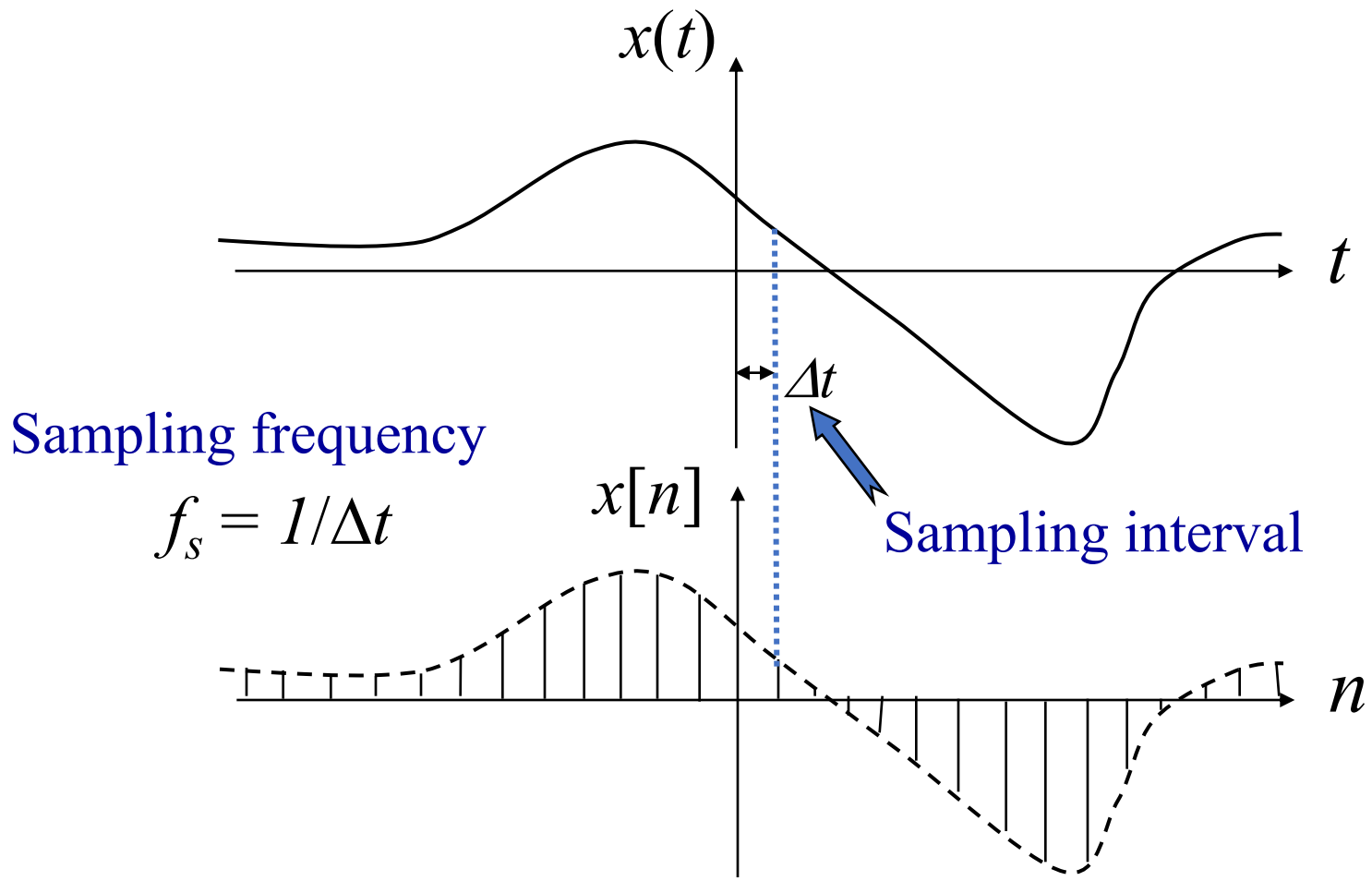
Digital representation converts a continuous amplitude into a number with specific quantization.

Example: +/- 1V with 4 levels

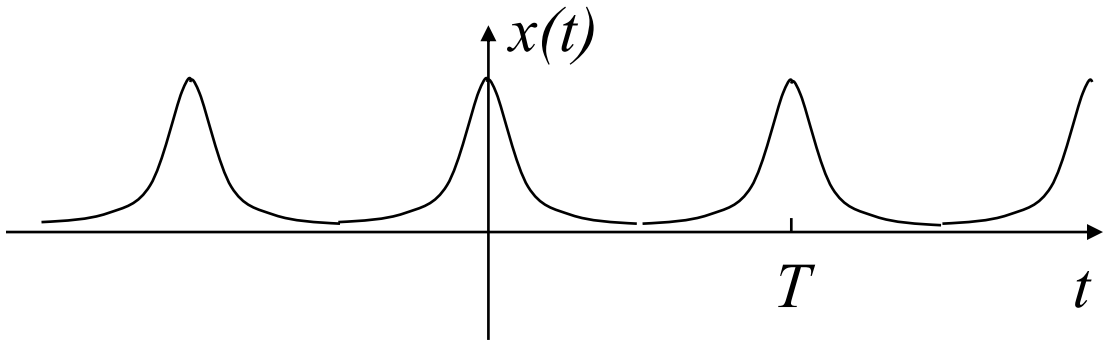
Continuous vs. Discrete Signals

Continuous

Discrete



Periodic vs. Non-Periodic Signals

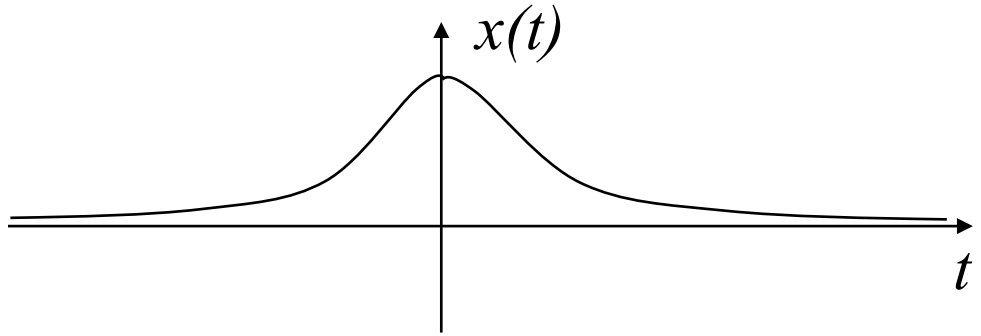


Periodic signal: $x(t)=x(t+T)$
(T is finite)

Period: $T, 2T, 3T,..NT\ldots$

⇓

Fundamental period



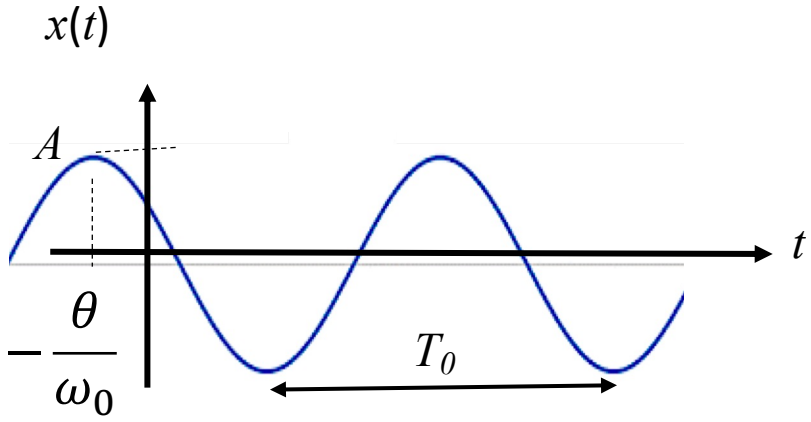
Aperiodic (non-periodic) signal
T is infinite

Special Signals

- Sinusoid
- Complex Exponential
- Unit Step
- Rectangular Pulse
- Impulse

Sinusoidal Signal

$$x(t) = A\cos(2\pi f_0 t + \theta)$$



A is a constant signal amplitude (voltage)

$2\pi f_0 = \omega_0$ is a constant signal frequency (ω in rad/s, f in Hz)

θ is a constant phase shift (rad)

$T_0 = \dfrac{1}{f_0}$ is the period

EULER'S THEOREM

$$e^{j\theta} = \cos\theta + j\sin\theta$$

$$\cos\theta = \tfrac{1}{2}\left(e^{j\theta} + e^{-j\theta}\right) \qquad \sin\theta = \tfrac{1}{2j}\left(e^{j\theta} - e^{-j\theta}\right) = \tfrac{-j}{2}\left(e^{j\theta} - e^{-j\theta}\right)$$

$$j = \sqrt{-1}$$

- Sinusoid can be written in exponential form

$$x(t) = Ae^{j(2\pi f_0 t + \theta)} = \mathrm{Re}\{x\} + \mathrm{Im}\{x\}$$
$$= A\cos(2\pi f_0 t + \theta) + jA\sin(2\pi f_0 t + \theta)$$

- Complex exponential – growing/decaying envelope & sinusoidal part

$$x(t) = Ae^{(\sigma_0 t + j(2\pi f_0 t + \theta))} = Ae^{\sigma_0 t} \quad e^{j(2\pi f_0 t + \theta)}$$

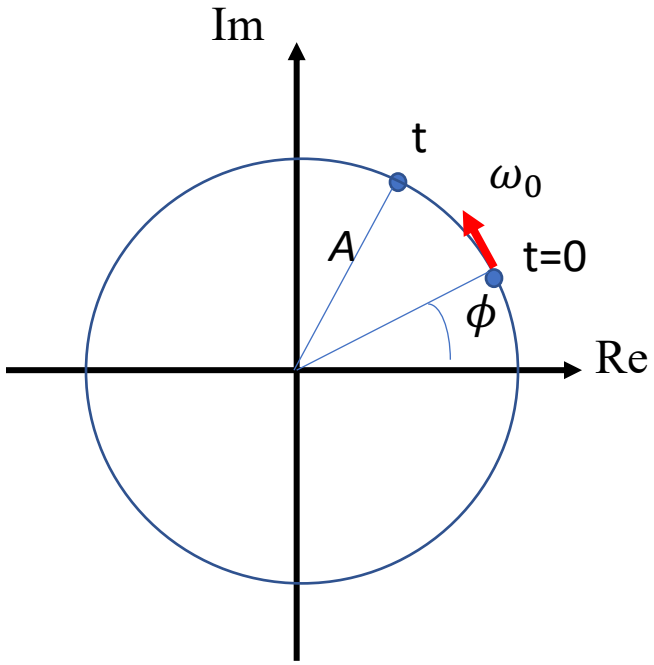
Complex Signals and Relationship with Sinusoids

$$x(t) = Ae^{j(\omega_0 t+\phi)}$$

Euler's formula: $e^{j\omega_0 t} = \cos\omega_0 t + j\sin\omega_0 t$

$$A\cos(\omega_0 t+\phi) = \frac{A}{2}(e^{j(\omega_0 t+\phi)} + e^{-j(\omega_0 t+\phi)})$$

$$A\sin(\omega_0 t+\phi) = \frac{A}{2j}(e^{j(\omega_0 t+\phi)} - e^{-j(\omega_0 t+\phi)})$$

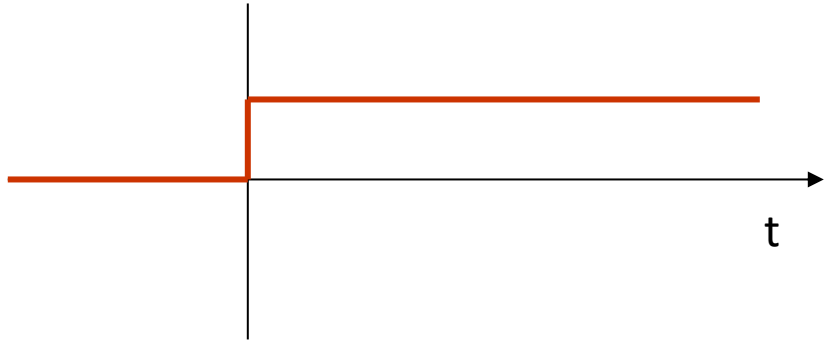


Both complex exponential and sinusoidal signals are periodic

SINGULARITY FUNCTIONS

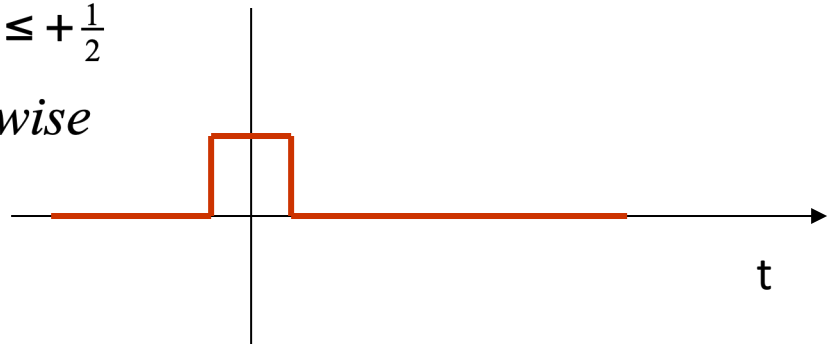
- Unit Step

$$u(t) \equiv \begin{cases} 1 & t \geq 0 \\ 0 & otherwise \end{cases}$$



- Unit Pulse

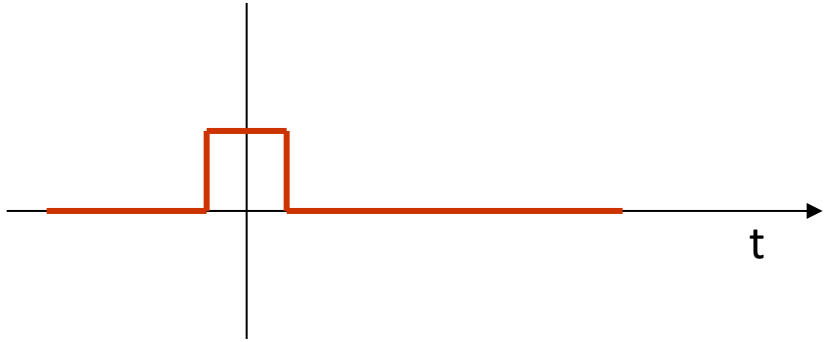
$$p(t) \equiv u\left(t+\tfrac{1}{2}\right) - u\left(t-\tfrac{1}{2}\right) = \begin{cases} 1 & -\tfrac{1}{2} \leq t \leq +\tfrac{1}{2} \\ 0 & otherwise \end{cases}$$



PULSES

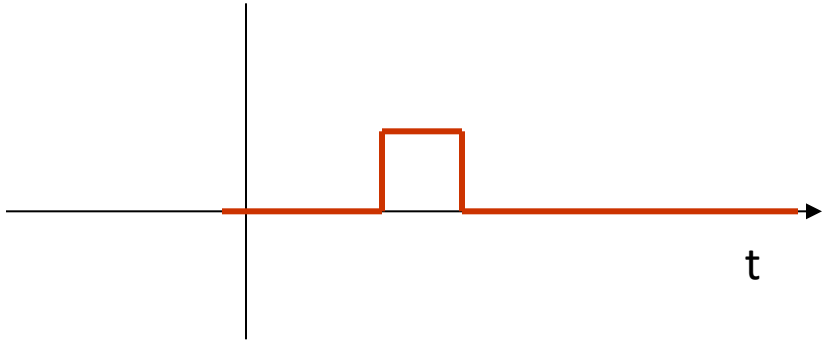
- Unit pulse

$$p(t) \equiv u\left(t+\tfrac{1}{2}\right) - u\left(t-\tfrac{1}{2}\right) = \begin{cases} 1 & -\tfrac{1}{2} \le t \le +\tfrac{1}{2} \\ 0 & otherwise \end{cases}$$



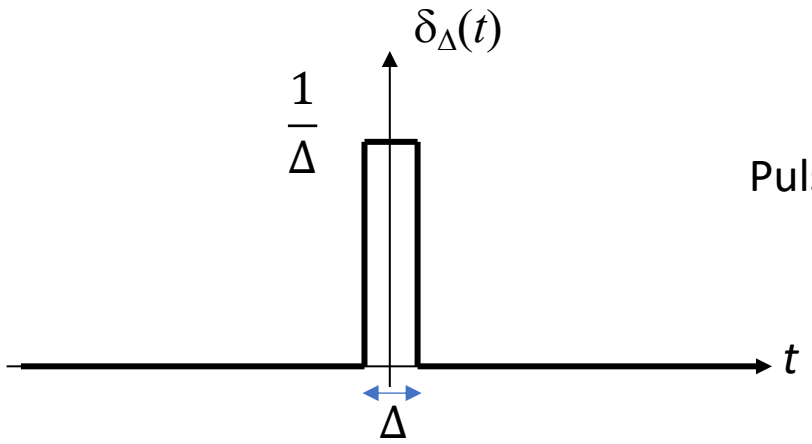
- Scaled and Shifted Pulse – unity area, width of T_c , offset of τ

$$\frac{1}{T_C} p\left(\frac{t-\tau}{T_C}\right)$$



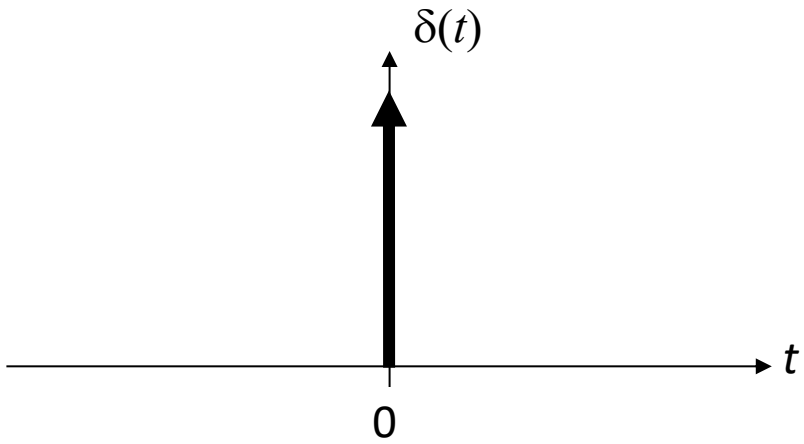
UNIT IMPULSE

Unit area, zero width



$\Delta \to 0$

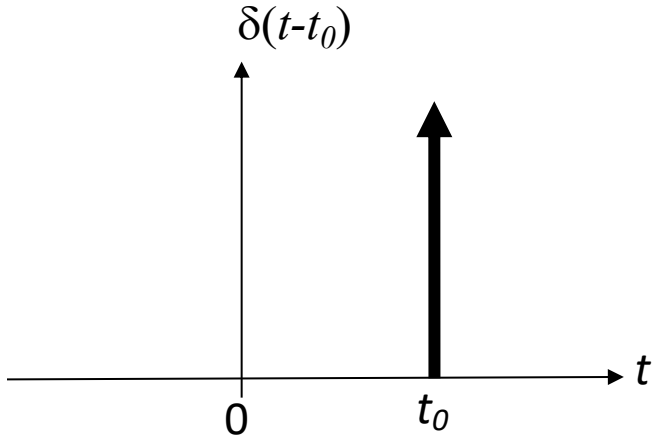
Pulse area = 1



$$\delta(t) = \lim_{\Delta \to 0} \delta_\Delta(t)$$

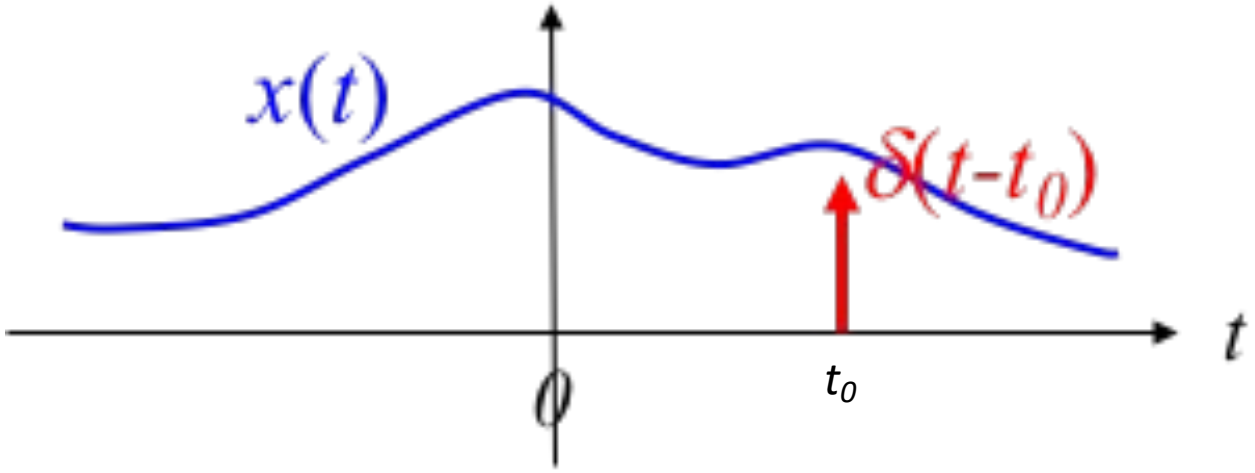
$$\delta(t) = \begin{cases} 0, t \neq 0 \\ undefined, \ otherwise \end{cases}$$

More general case:



$$\delta(t - t_0) = \begin{cases} 0, t \neq t_0 \\ undefined, \ otherwise \end{cases}$$

Sampling Property of Unit Impulse: Continuous

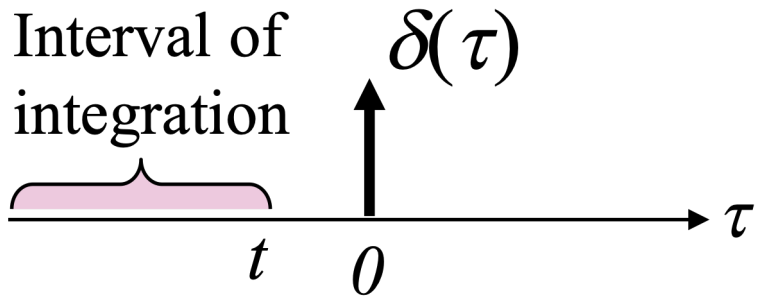


$$x(t)\delta(t-t_0) = x(t_0)\delta(t-t_0)$$

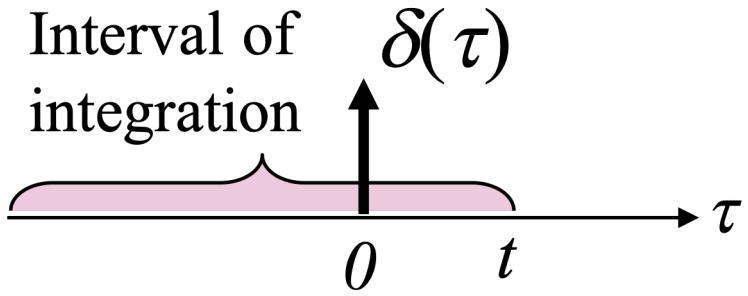
A special case: $x(t)\delta(t) = x(0)\delta(t)$

Unit Impulse & Step Relationship: Continuous time

$$u(t) = \int_{-\infty}^{t} \delta(\tau) d\tau \qquad \delta(t) = \frac{du(t)}{dt}$$

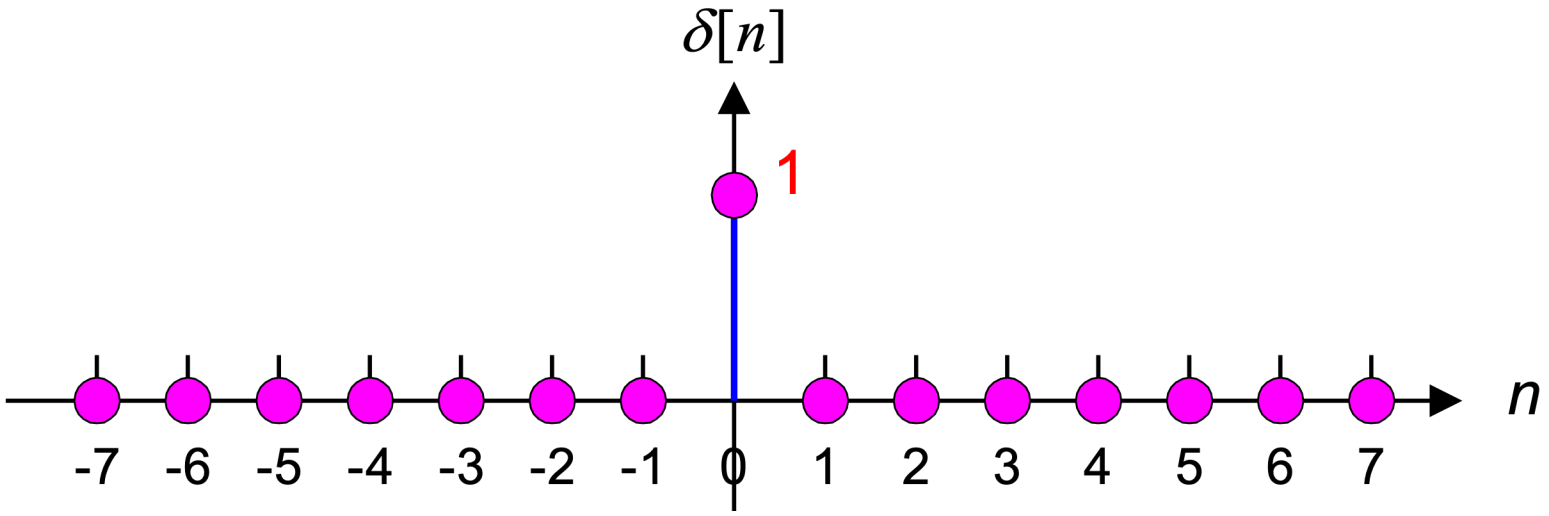


$t < 0, u(t) = 0$

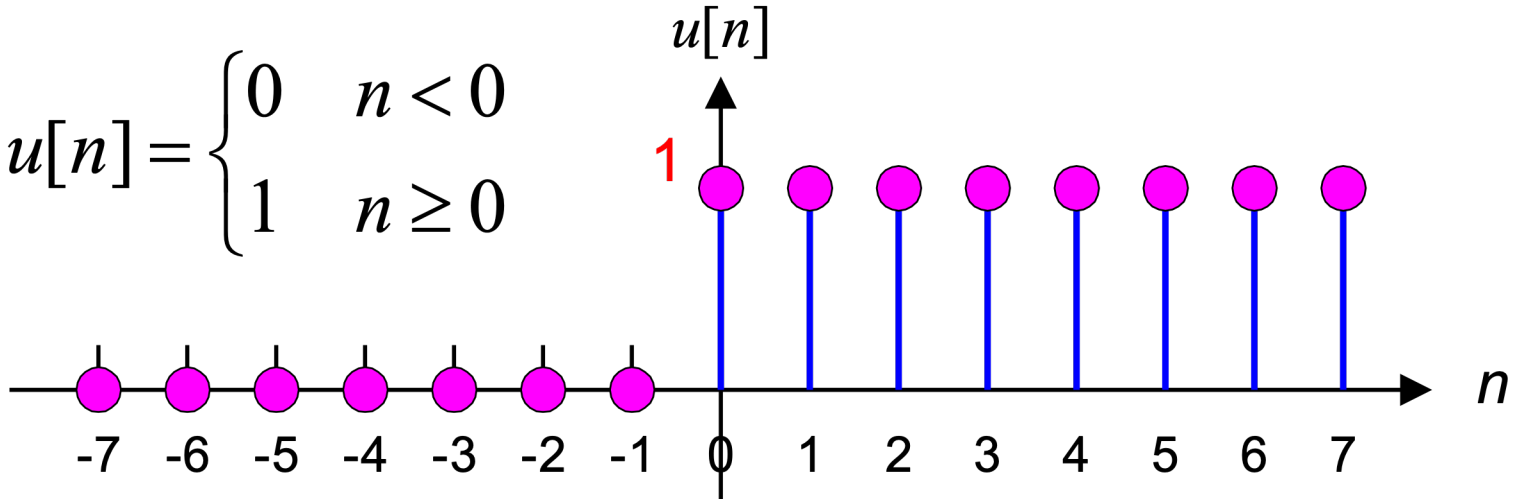


$t > 0, u(t) = 1$

Unit Impulse and Step Function: Discrete



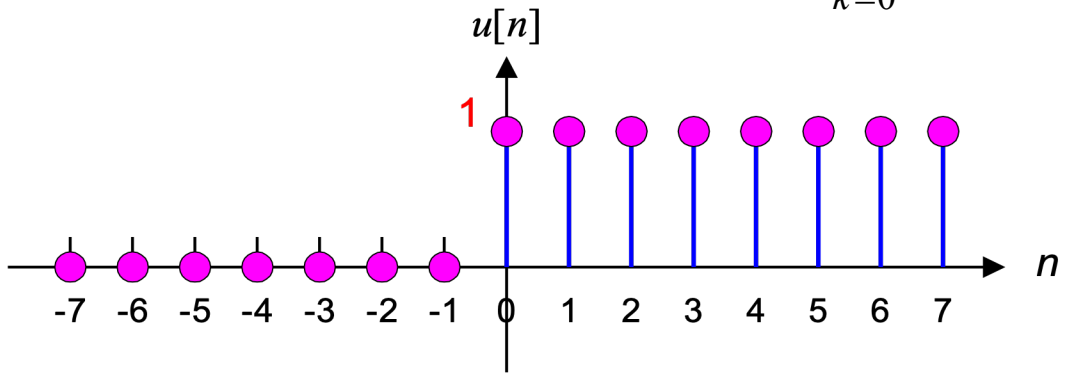
$$u[n] = \begin{cases} 0 & n < 0 \\ 1 & n \geq 0 \end{cases}$$

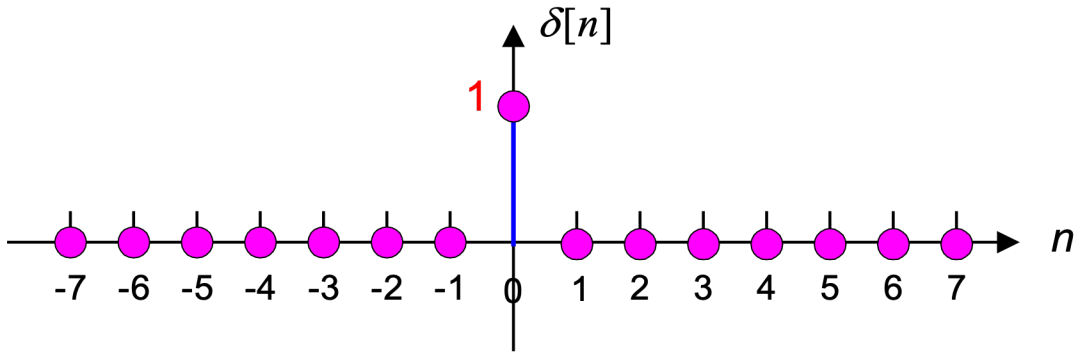


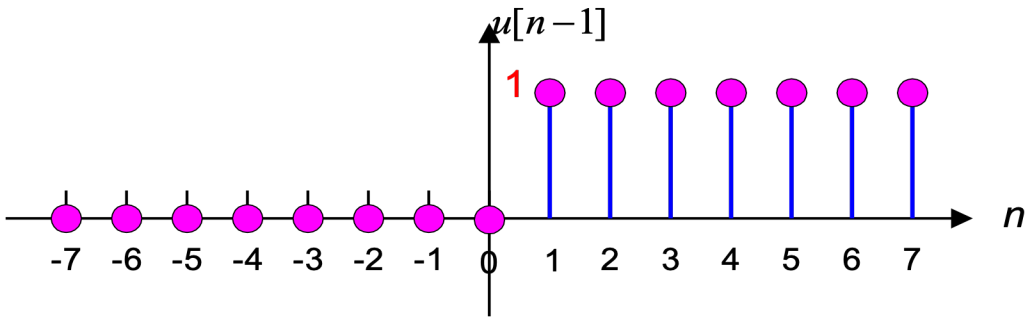
Discrete Impulse & Step Function Relationship

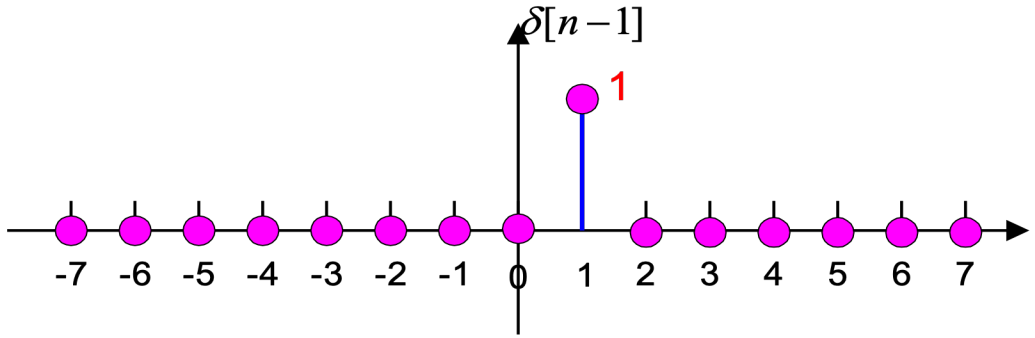
$$u[n] = \sum_{k=0}^{\infty} \delta[n-k]$$

$$\delta[n] = u[n] - u[n-1]$$

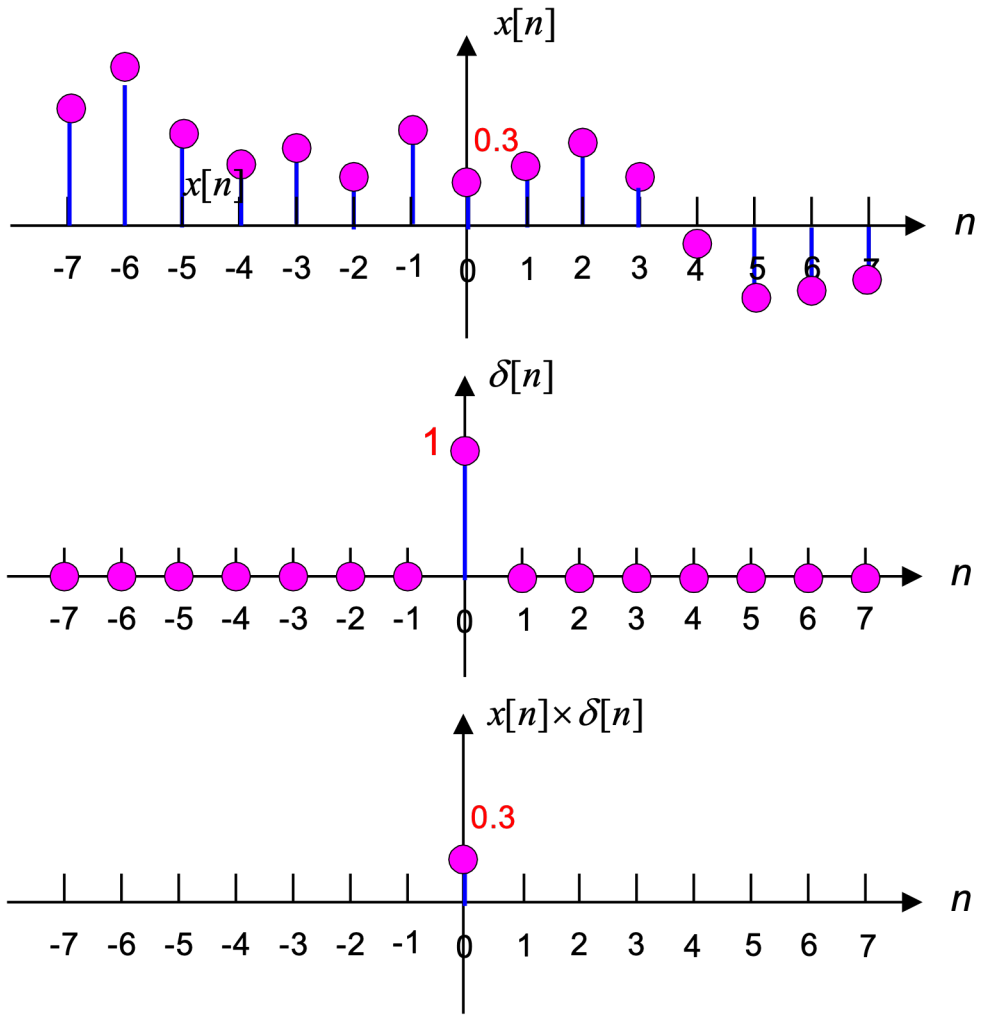




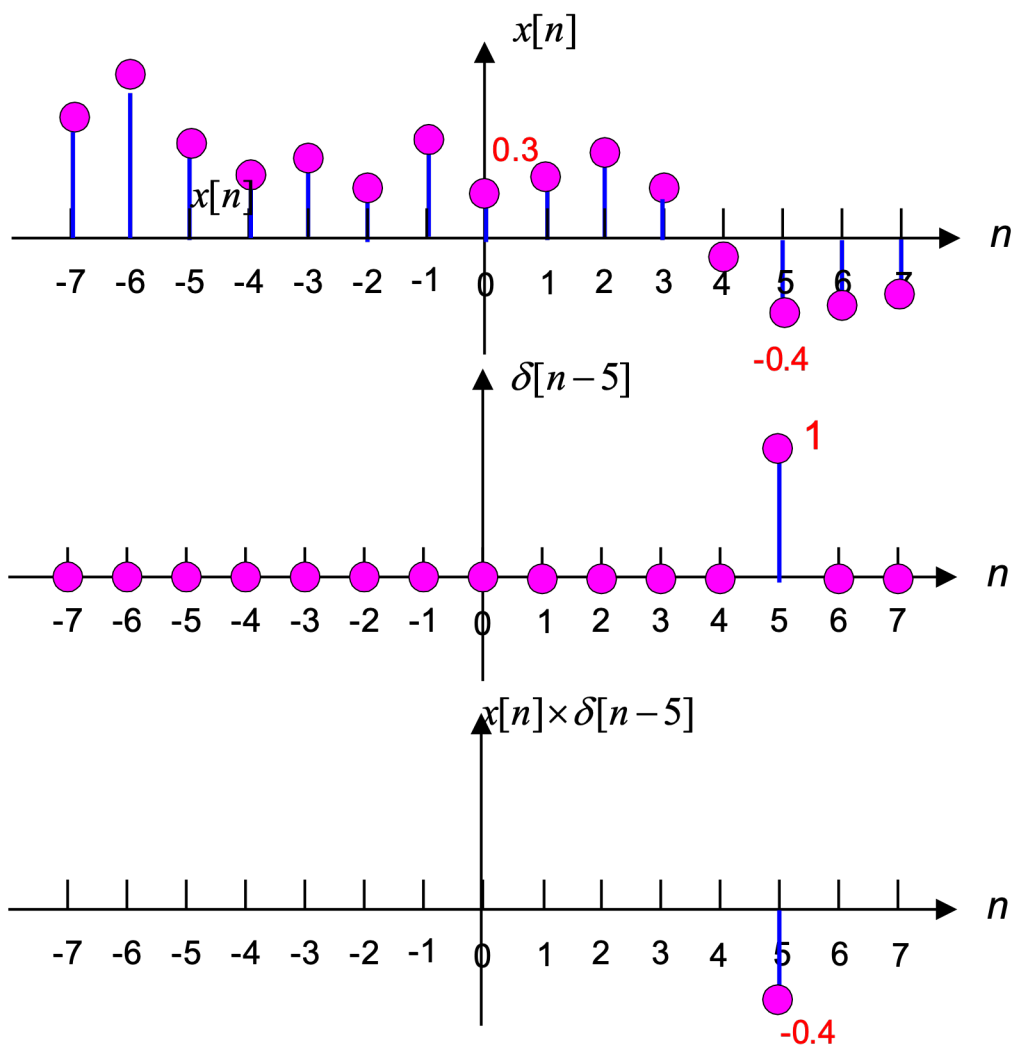




Discrete Impulse Sampling Property



$$x[n]\delta[n] = x[0]$$



$$x[n]\delta[n-k] = x[k]$$

Signal Energy and Power

Energy/Power	Continuous	Discrete
Energy	$E = \int_{-\infty}^{\infty} \lvert x(t) \rvert^2 \, dt$	$E = \sum_{n=-\infty}^{\infty} \lvert x[n] \rvert^2$
Average Power	$P = \frac{1}{T} \int_{-T/2}^{T/2} \lvert x(t) \rvert^2 \, dt$	$P = \frac{1}{2N+1} \sum_{n=-N}^{N} \lvert x[n] \rvert^2$

(T→∞, N→∞, for non-periodic signals)

Exercise: compute the power and energy for the following signals:

$x(t) = u(t)$ $\qquad$ $x(t) = e^{j(2t+\pi/4)}$ $\qquad$ $x[n] = \cos\left(\frac{\pi}{4} n\right)$

Signal Energy

- Energy in a signal (assuming R=1 ohm)

$$\varepsilon = \int_{t_1}^{t_2} |x(t)|^2 dt$$

- Example for a unit area pulse

$$\varepsilon\left\{\frac{1}{T_C} p\left(\frac{t-\tau}{T_C}\right)\right\} = \int_{-\infty}^{+\infty} \left|\frac{1}{T_C} p\left(\frac{t-\tau}{T_C}\right)\right|^2 dt = \frac{1}{{T_C}^2} \int_{\tau-\frac{T_C}{2}}^{\tau+\frac{T_C}{2}} (1)^2 dt = \frac{1}{T_C}$$

SIGNAL POWER

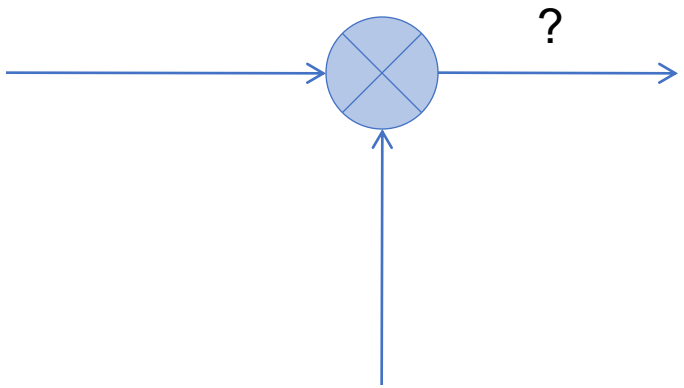
- Time averaged power

$$P = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} \left| x(t) \right|^2 dt \quad \text{(watts)}$$

- Example for a sinusoid

$$P = \lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} \left| A\cos\left(2\pi f_0 t + \theta\right) \right|^2 dt \quad = \frac{A^2}{2}$$

MIXING or MULTIPLYING SINUSOIDAL SIGNALS

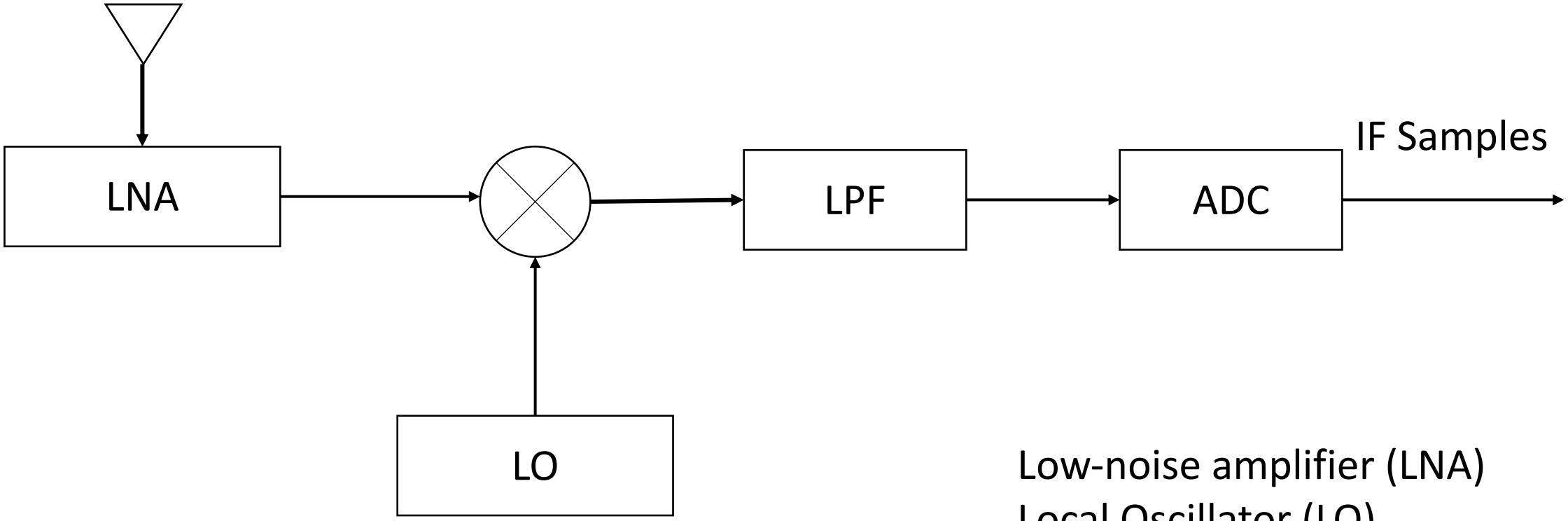


$$sin^2x + cos^2x = 1$$

$$\cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y)$$

$$\cos(x)\cos(y) = \frac{1}{2}\cos(x+y) + \frac{1}{2}\cos(x-y)$$

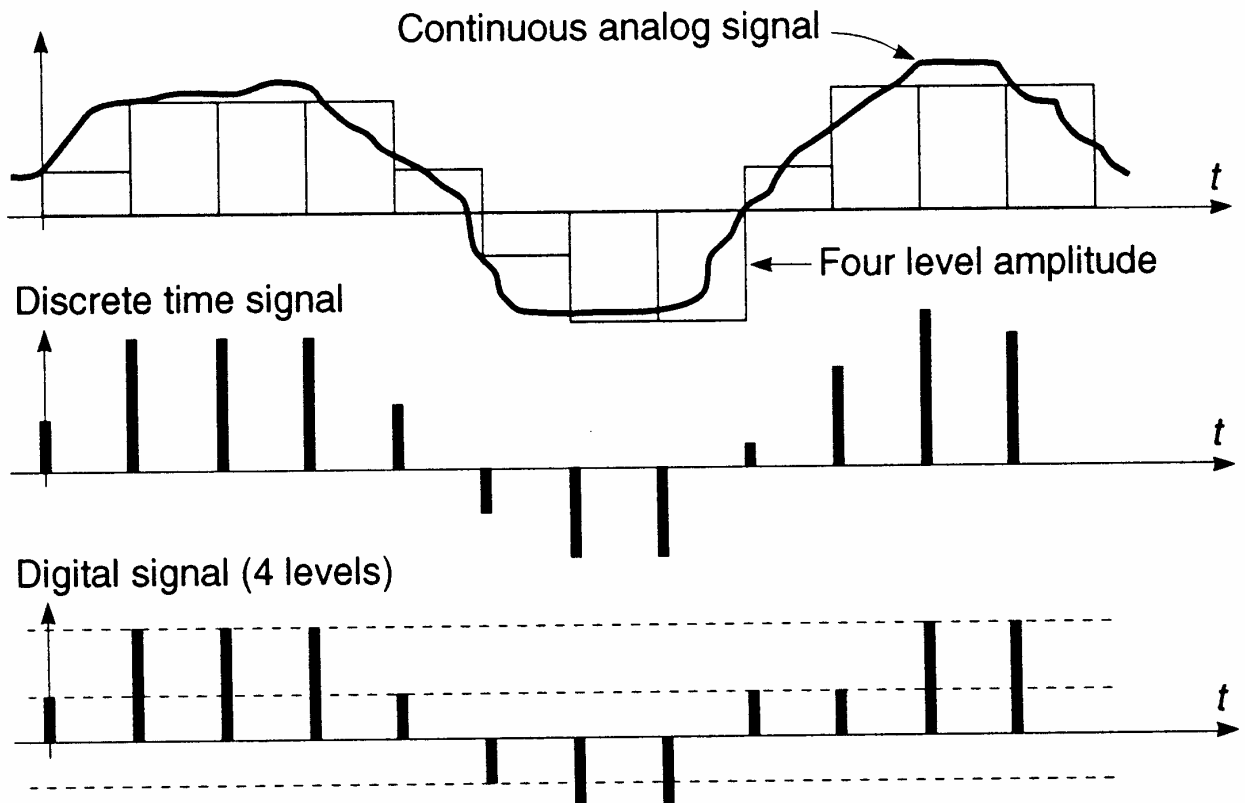
Up/down conversion



Low-noise amplifier (LNA)
Local Oscillator (LO)
Low-pass filter (LPF)
Intermediate Frequency (IF)

A/D Conversion

GLOBAL POSITIONING SYSTEM



Sampling frequency and aliasing

- Baseband sampling - to accurately capture a signal one must sample at least twice as fast as the highest frequency component of the signal. (Nyquist rate)

- Bandpass sampling – if the original signal is filtered to a specified bandwidth, one can sample at a lower frequency such that the signal uniquely aliases to baseband – this combines sampling and downconversion in one step

Aliasing

- To find the frequency that will be seen with a sampling frequency fs,
  - Reflect or fold around fs/2 if positive: f -> fs-f
  - Reflect or fold around 0 if negative: -f -> +f
- Examples: Say fs = 10 Hz, fs/2 = 5 Hz
  - f1=3Hz -> 3Hz (no change)
  - f2=6Hz -> 4Hz (reflects around fs/2)
  - f3=10Hz -> 0 Hz
  - f4=12Hz -> -2Hz -> +2Hz
  - f5=17Hz -> -7Hz -> +7Hz -> 3 Hz
- Also check out the “image ladder” in the textbook, Fig 11.13

Bandpass sampling – deliberate aliasing

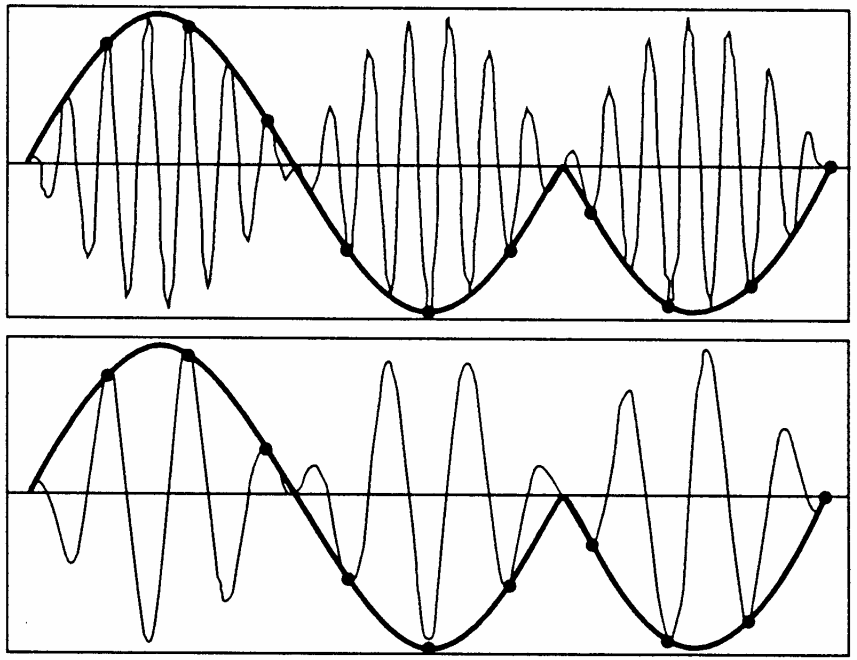


Figure 9.13 Bandpass sampling of a desired signal and an alias.

Analog to Digital Conversion

- Sampling frequency must be higher than Nyquist
  – what is required for C/A? For P(Y)?
- 1 bit A/D converters often used in low cost GPS
  - Why does this work?
  - Some loss of signal but not much
- 2 bit requires use of Automatic Gain Control (AGC) to keep samples well-distributed
- Higher resolution not very useful for many applications

Periodic Functions

- A periodic function of period T_p = 1/fp

$$f\left(t+nT_p\right)=f\left(t\right)$$

- Can be expressed by a Fourier Trigonometric Series as:

$$f\left(t\right)=a_0+\sum_{n=1}^{\infty}\left[a_n\cos n\left(2\pi f_p t\right)+b_n\sin n\left(2\pi f_p t\right)\right]$$

Synthesis equation – shows how f(t) can be reconstructed by adding up an infinite number of sines & cosines

Fourier series for a signal that is periodic in P

$$f(t) = a_0 + \sum_{n=1}^{\infty}\left[a_n \cos n\left(2\pi f_p t\right) + b_n \sin n\left(2\pi f_p t\right)\right]$$

analysis equations – show how to find the amount of each sine and cosine wave that is needed.

$$a_0 = \frac{1}{P}\int_{-P/2}^{P/2} f(t)\,dt$$

$$a_n = \frac{1}{P/2}\int_{-P/2}^{P/2} f(t)\cos n2\pi f_p t\;dt$$

$$b_n = \frac{1}{P/2}\int_{-P/2}^{P/2} f(t)\sin n2\pi f_p t\;dt$$

Change of variable

Let $x = 2\pi f_p t,$

so for $t = T_p, x = 2\pi$

The function f is then expressed as:

$$f(x) = a_0 + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx]$$

Example:

$f(x) = 12sinx - 3cos2x$

What should the coefficients be?

b1 = 12, a2 = -3, all the rest = 0

Fourier Coefficients

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\, dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx\ dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx\ dx$$

Examples

$$f(x) = a_0 + \sum_{n=1}^{\infty}\left[a_n \cos nx + b_n \sin nx\right]$$

$f(x)$	a_0	a_n	b_n
Constant (A)	$\frac{1}{2\pi}\int_{-\pi}^{\pi} A\,dx = A$	$\frac{1}{\pi}\int_{-\pi}^{\pi} A\cos nx\ dx = 0$	$\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\ dx = 0$
$\cos nx$	$\frac{1}{2\pi}\int_{-\pi}^{\pi} \cos nx\ dx = 0$	$\frac{1}{\pi}\int_{-\pi}^{\pi} \cos^2 nx\ dx = 1$	$\frac{1}{\pi}\int_{-\pi}^{\pi} \cos nx\ \sin nx\ dx = 0$
$\cos mx \quad (m \neq n)$	$\frac{1}{2\pi}\int_{-\pi}^{\pi} \cos mx\ dx = 0$	$\frac{1}{\pi}\int_{-\pi}^{\pi} \cos mx \cos nx\ dx = 0$	$\frac{1}{\pi}\int_{-\pi}^{\pi} \cos mx\ \sin nx\ dx = 0$
$\sin nx$	$\frac{1}{2\pi}\int_{-\pi}^{\pi} \sin nx\ dx = 0$	$\frac{1}{\pi}\int_{-\pi}^{\pi} \sin nx\ \cos nx\ dx = 0$	$\frac{1}{\pi}\int_{-\pi}^{\pi} \sin^2 nx\ dx = 1$

Example

Example:

$$f(x) = 12sinx - 3cos2x$$

What should the coefficients be?
b1 = 12, a2 = -3, all the rest = 0

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\;dx$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\;dx$$

$$sin^2x + cos^2x = 1$$

$$\cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y)$$

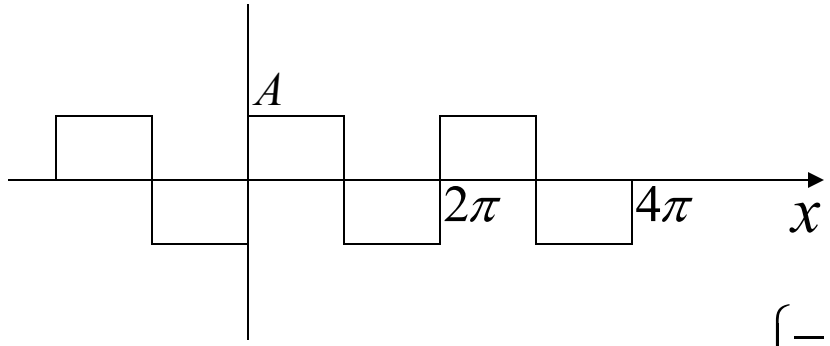
$$\cos(x)\cos(y) = \frac{1}{2}\cos(x+y) + \frac{1}{2}\cos(x-y)$$

$$\sin(x)\cos(x) = \frac{1}{2}\sin 2x$$

$$cos^2(x) = \frac{1}{2}(1+\cos 2x)$$

$$sin^2(x) = \frac{1}{2}(1-\cos 2x)$$

Square Wave



$$f(x) = \begin{cases} -A & \text{if } -\pi < x < 0 \\ A & \text{if } 0 < x < \pi \end{cases}$$

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{0} -A\,dx + \frac{1}{2\pi}\int_{0}^{\pi} A\,dx = 0$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{0} -A\cos nx\;dx + \frac{1}{\pi}\int_{0}^{\pi} A\cos nx\;dx = 0 \text{ for all } n$$

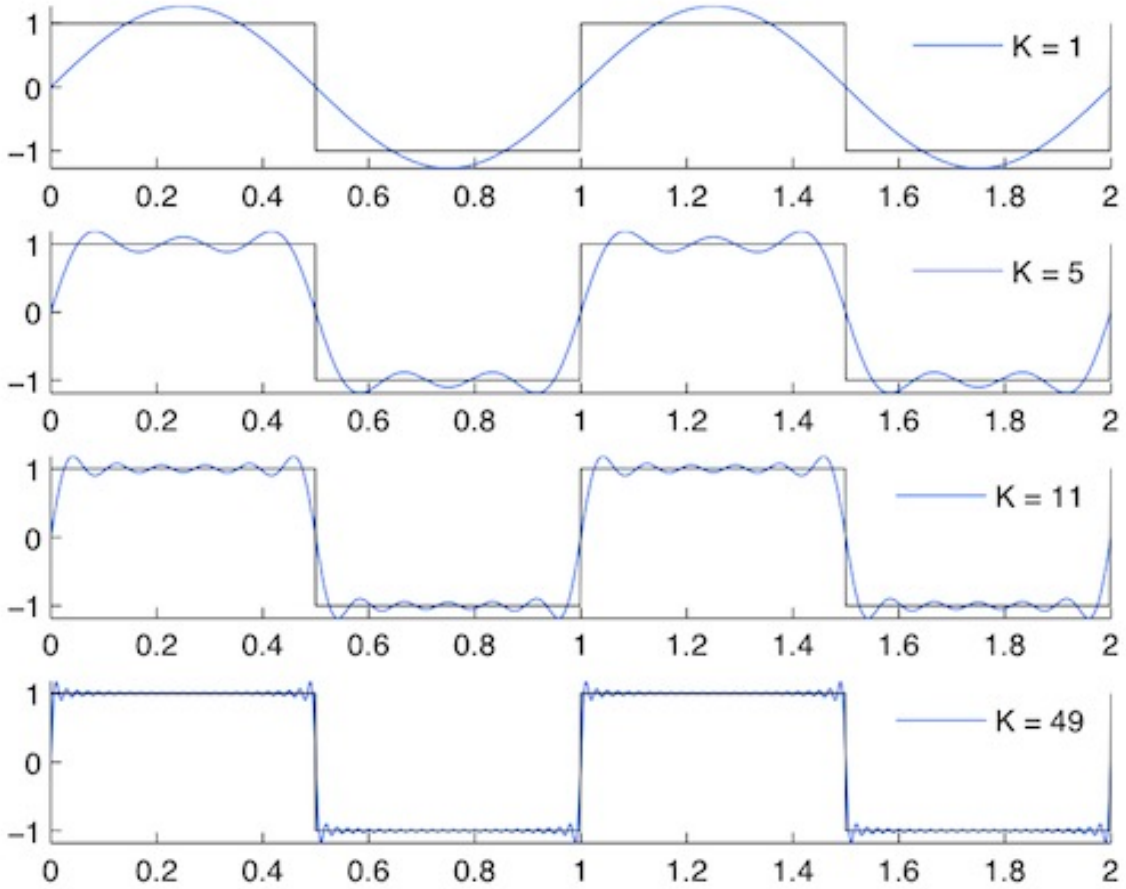
$$b_n = \frac{1}{\pi}\int_{-\pi}^{0} -A\sin nx\;dx + \frac{1}{\pi}\int_{0}^{\pi} A\sin nx\;dx = \frac{2A}{n\pi}(1-\cos n\pi)$$

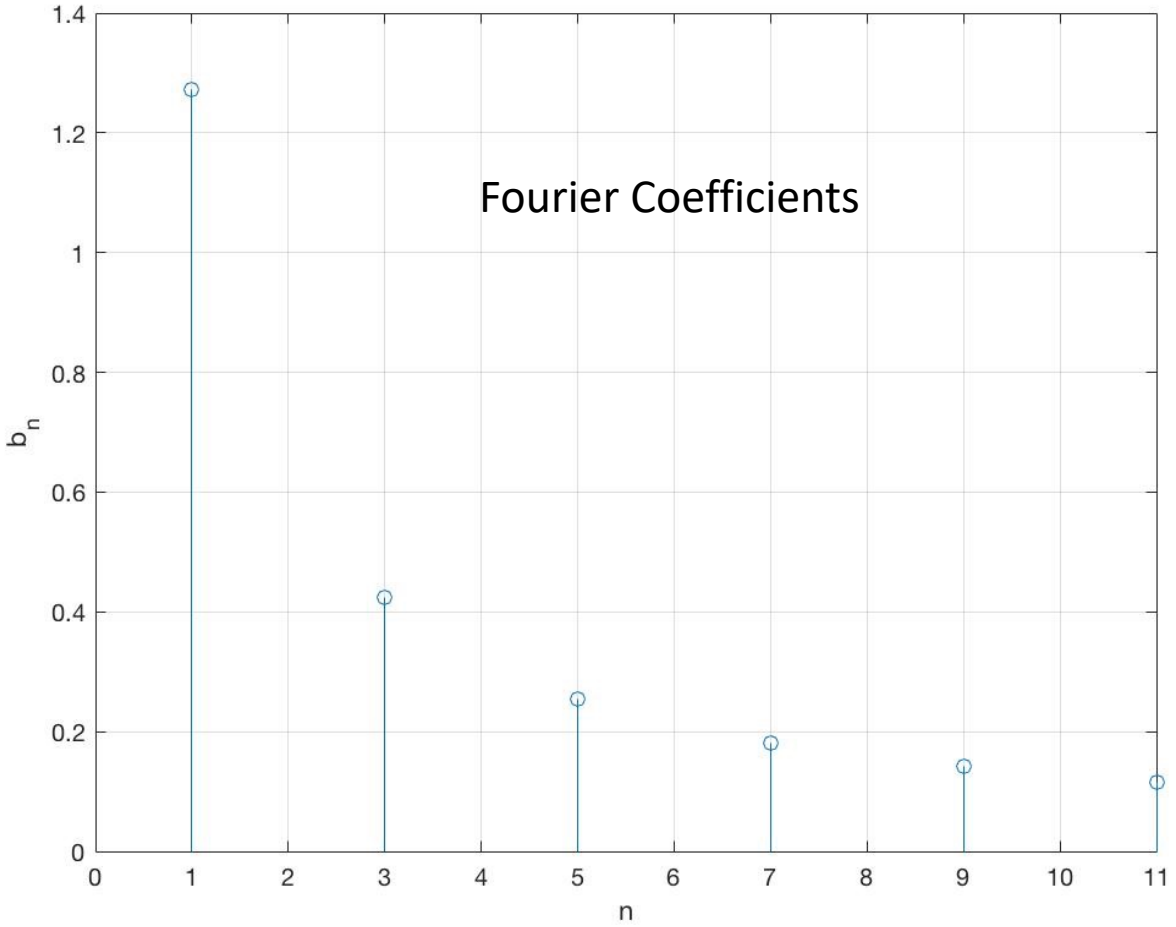
$$b_1 = \frac{4A}{\pi}\,,\; b_2 = 0\,,\; b_3 = \frac{4A}{3\pi}\,,\; b_4 = 0\,,etc.$$

$$\text{so}\quad f(x) = \frac{4A}{\pi}\left[\sin x + \tfrac{1}{3}\sin 3x + \tfrac{1}{5}\sin 5x + \ldots\right]$$

Reconstructed function based on the Fourier series

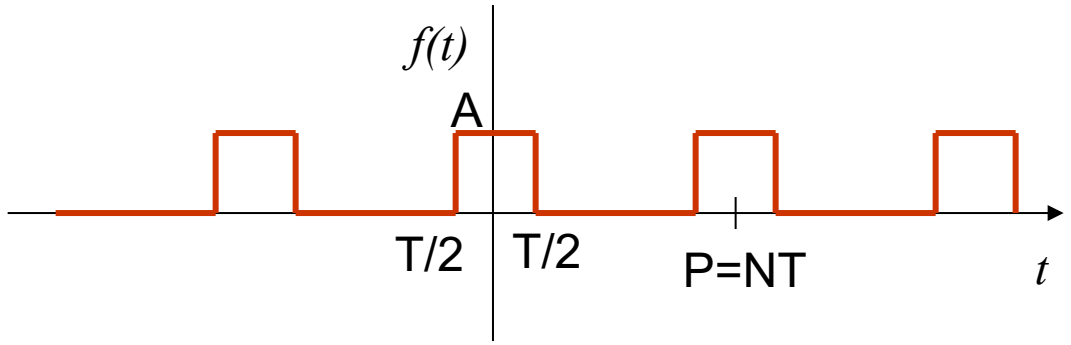
Fourier Series Approximations





Periodic rectangular pulse

$$f(t) = \begin{cases} 0 & \text{if } |t| > \frac{T}{2} \\ A & \text{if } |t| < \frac{T}{2} \end{cases}$$



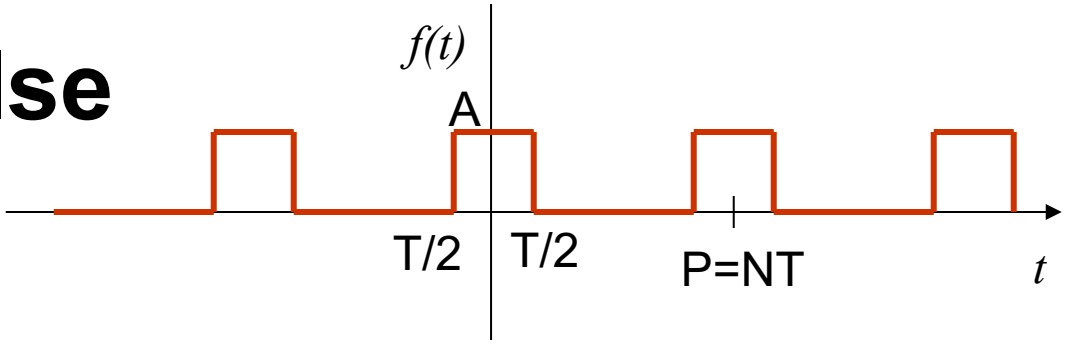
The textbook refers to this a square wave in Fig 8.12.

$$P = NT = \frac{1}{f_P}$$

How can we express as a Fourier trig series?

$$f(t) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos n\left(2\pi f_p t\right) + b_n \sin n\left(2\pi f_p t\right) \right]$$

Periodic rectangular pulse

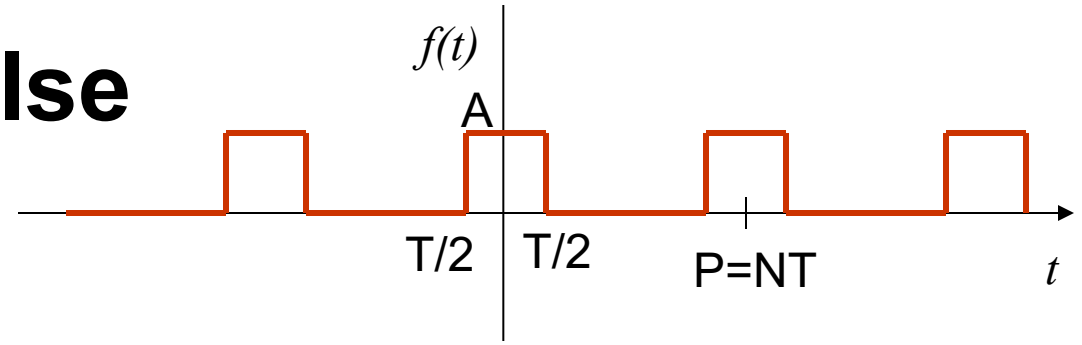


$$f(t) = \begin{cases} 0 & \text{if } |t| > \frac{T}{2} \\ A & \text{if } |t| < \frac{T}{2} \end{cases}$$

$$P = NT = \frac{1}{f_P}$$

$$a_0 = \frac{1}{P}\int_{-P/2}^{P/2} f(t)\,dt = \frac{1}{P}\int_{-T/2}^{T/2} A\,dt = \frac{AT}{P} = \frac{A}{N}$$

Periodic rectangular pulse



$$f(t) = \begin{cases} 0 & \text{if } |t| > \frac{T}{2} \\ A & \text{if } |t| < \frac{T}{2} \end{cases} \qquad P = NT = \frac{1}{f_P}$$

$$a_n = \frac{1}{P/2}\int_{-P/2}^{P/2} f(t)\cos n2\pi f_p t \; dt = \frac{1}{P/2}\int_{-T/2}^{T/2} A\cos n2\pi f_p t \; dt = \frac{A\sin n2\pi f_p t}{n2\pi f_p P/2}\Bigg|_{T/2}^{T/2}$$

How does this compare to square wave?

$$a_n = \frac{2A\sin n\pi f_p T}{n\pi} = \frac{2A}{N}\frac{\sin n\pi/N}{n\pi/N} = \frac{2A}{N}\operatorname{sinc}\frac{n\pi}{N}$$

$$a_1 = \frac{2A}{N}\frac{\sin \pi/N}{\pi/N}, a_2 = \frac{2A}{N}\frac{\sin 2\pi/N}{2\pi/N}, \ldots$$

Remember what these coefficients mean:

$$f(t) = a_0 + \sum_{n=1}^{\infty}\left[a_n \cos n\left(2\pi f_p t\right) + b_n \sin n\left(2\pi f_p t\right)\right]$$

Periodic rectangular pulse (continued)

For what values of "n" is a_n=0? (what multiples of the repeat frequency?)

$$a_n = \frac{2A}{N}\frac{\sin n\pi / N}{n\pi / N} = 0 \text{ for } \quad \frac{n}{N} = \text{is a non-zero integer}$$

$$n = N, 2N, 3N, etc$$

$$Nf_P = \frac{1}{T} = f_C$$

So, for $Nf_P t,\ 2Nf_P t,\ 3Nf_P t,\ 4Nf_P t$

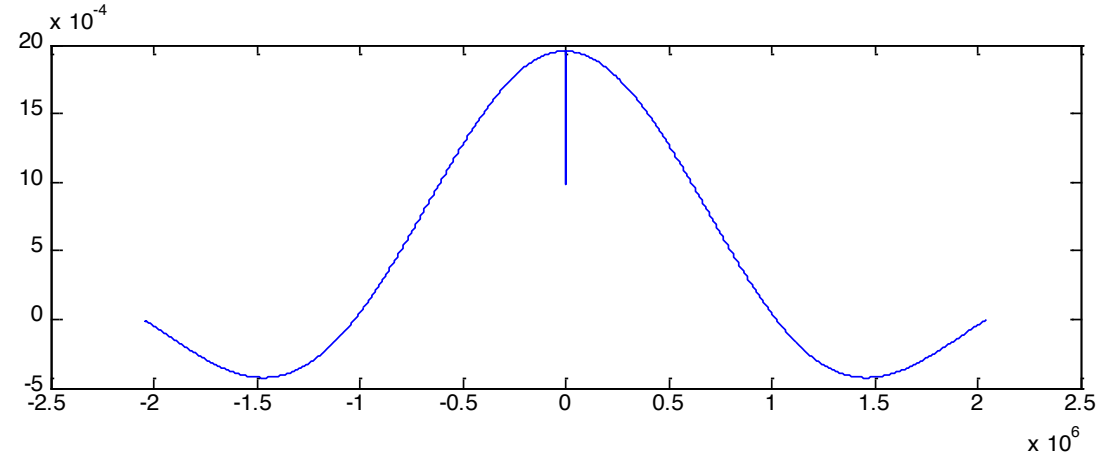
and $\quad -Nf_P t,\ -2Nf_P t,\ -3Nf_P t,\ -4Nf_P t$

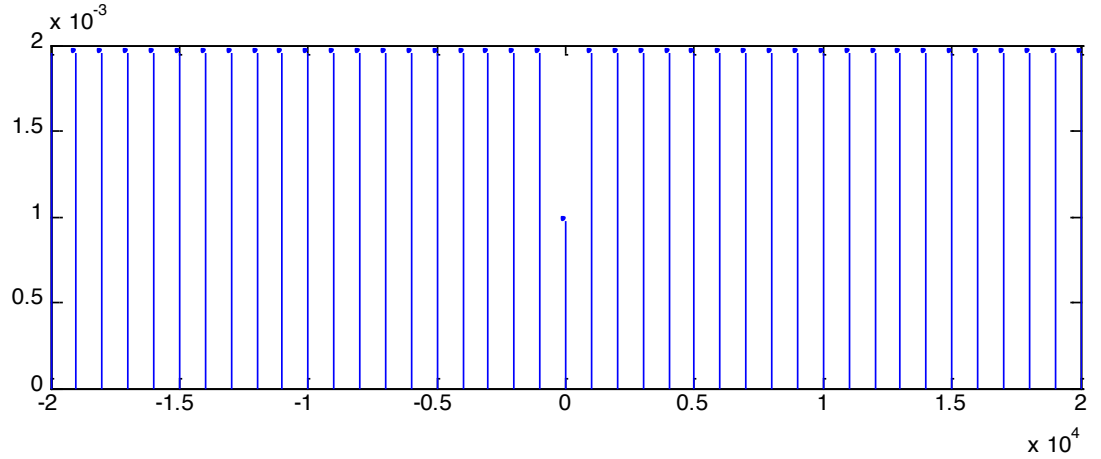
$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\left(2\pi f_p t\right)$$

Fourier representation of a periodic rectangular pulse

$$f(t) = \frac{A}{N} + \frac{2A}{N}\sum_{n=1}^{\infty}\left[\operatorname{sinc}\frac{n\pi}{N}\cos n\left(2\pi f_p t\right)\right]$$

a_n





Series for 1/T=1.023e6 and N = 1023

Repeat period of signal = 0.001 sec

f_P = 1000 Hz

Zoomed in to show individual values

FOURIER TRANSFORM

- For any $x(t)$ you can take a Fourier Transform, to produce a continuous function of frequency:

$$X(f) = \mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}dt$$

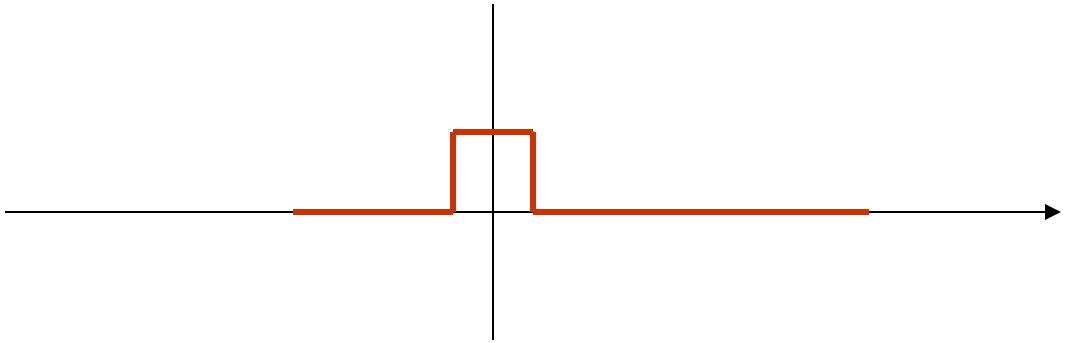
- To go from frequency domain to time domain use the inverse transform:

$$x(t) = \mathcal{F}^{-1}\{X(f)\} = \int_{-\infty}^{\infty} X(f)e^{j2\pi ft}df$$

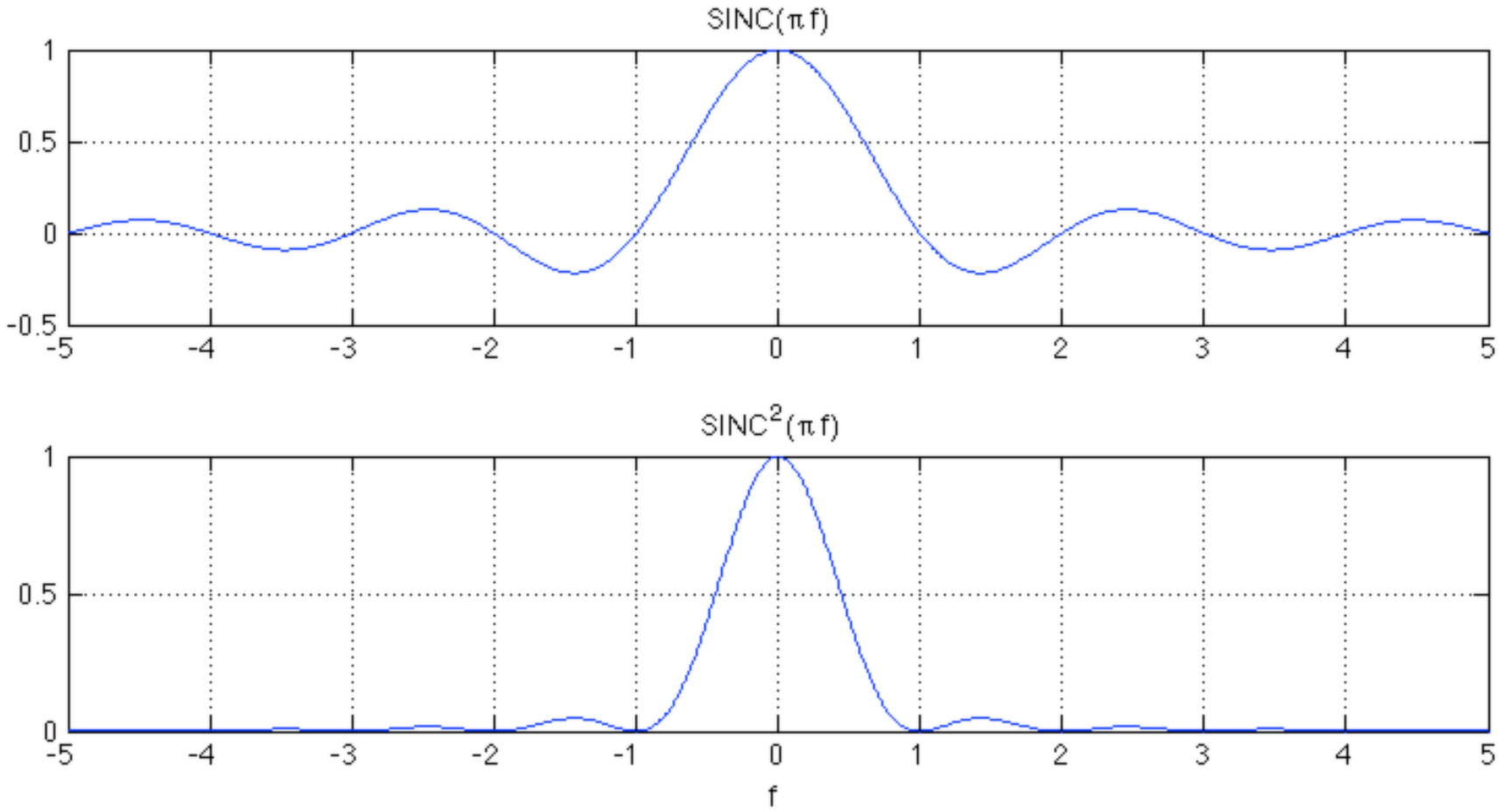
Table 8.2 Fourier transform pairs

Time function	*Fourier transform*
Unit pulse $p(t)$	$\text{sinc}(\pi f)$
Shifted and scaled pulse $Ap\left(\frac{t-\tau}{T_C}\right)$	$AT\exp(-j2\pi f\tau)\text{sinc}(\pi fT)$
Unit impulse function $\delta(t)$	1
$\delta(t-t_0)$	$\exp(-j2\pi ft_0)$
1	$\delta(f)$
$\exp(j2\pi f_0 t)$	$\delta(f-f_0)$
$\cos(2\pi f_0 t)$	$\frac{1}{2}\delta(f-f_0)+\frac{1}{2}\delta(f+f_0)$
$\sin(2\pi f_0 t)$	$\frac{1}{2j}\delta(f-f_0)-\frac{1}{2j}\delta(f+f_0)$
$\text{sgn}\,t=\begin{cases}1 & t\geq 0\\ -1 & t<0\end{cases}$	$\frac{1}{j\pi f}$
Step function $u(t)$	$\frac{1}{2}\delta(f)+\frac{1}{j2\pi f}$
Ramp function $tu(t)$	$j\pi\dot{\delta}(f)-\frac{1}{(2\pi f)^2}$
Comb function $\sum_{i=-\infty}^{\infty}\delta(t-iT_S)$	$\frac{1}{T_S}\sum_{i=-\infty}^{\infty}\delta\left(f-\frac{i}{T_S}\right)$

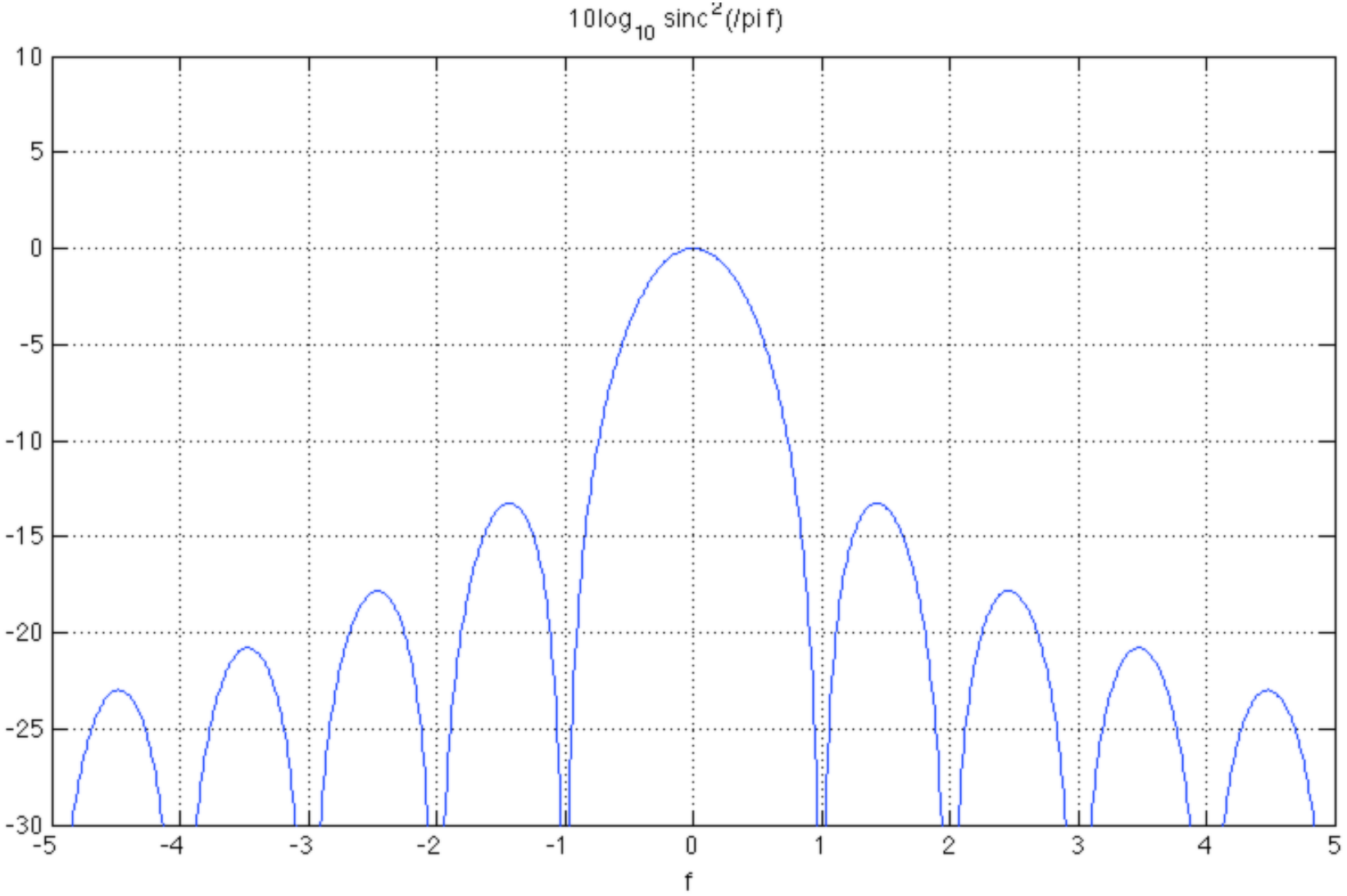
Fourier Transform of a Unit Pulse



Sinc and Sinc^2



sinc^2 in dB



Fourier Transform Properties

Table 8.1 Properties of Fourier transforms

Property/operation	*Time function*	*Fourier transform*
Linearity	$k_1 x(t) + k_2 y(t)$	$k_1 X(f) + k_2 Y(f)$
Time delay	$x(t - t_0)$	$X(f)\exp(-j2\pi f t_0)$
Time scaling	$x(kt)$	$\frac{1}{\lvert k\rvert} X\left(\frac{f}{k}\right)$
Differentiation	$\frac{dx}{dt}$	$j2\pi f X(f)$
n differentiations	$\frac{d^n x}{dt^n}$	$(j2\pi f)^n X(f)$
Symmetry	$\mathcal{F}\{x(t)\} = X(f)$	$\mathcal{F}\{X(t)\} = x(-f)$
Frequency translation	$x(t)\exp(j2\pi f_0 t)$	$X(f - f_0)$
Modulation	$x(t)\cos(j2\pi f_0 t)$	$\frac{1}{2} X(f - f_0) + \frac{1}{2} X(f + f_0)$
Modulation	$x(t)\sin(j2\pi f_0 t)$	$\frac{1}{2j} X(f - f_0) - \frac{1}{2j} X(f + f_0)$
Convolution	$x(t) * h(t)$	$X(f)H(f)$
Multiplication	$x(t) \times h(t)$	$X(f) * H(f)$
Integration	$\int_{-\infty}^{t} x(\varphi)\, d\varphi$	$\frac{X(f)}{j2\pi f} + \frac{1}{2} X(0)\delta(f)$

Fourier Transform Frequency Shift

Start with a signal centered on f_{L1}

$$\Im\{a(t)\cos 2\pi f_0 t\} = \frac{1}{2}A(f - f_0) + \frac{1}{2}A(f + f_0) \qquad (9.3)$$

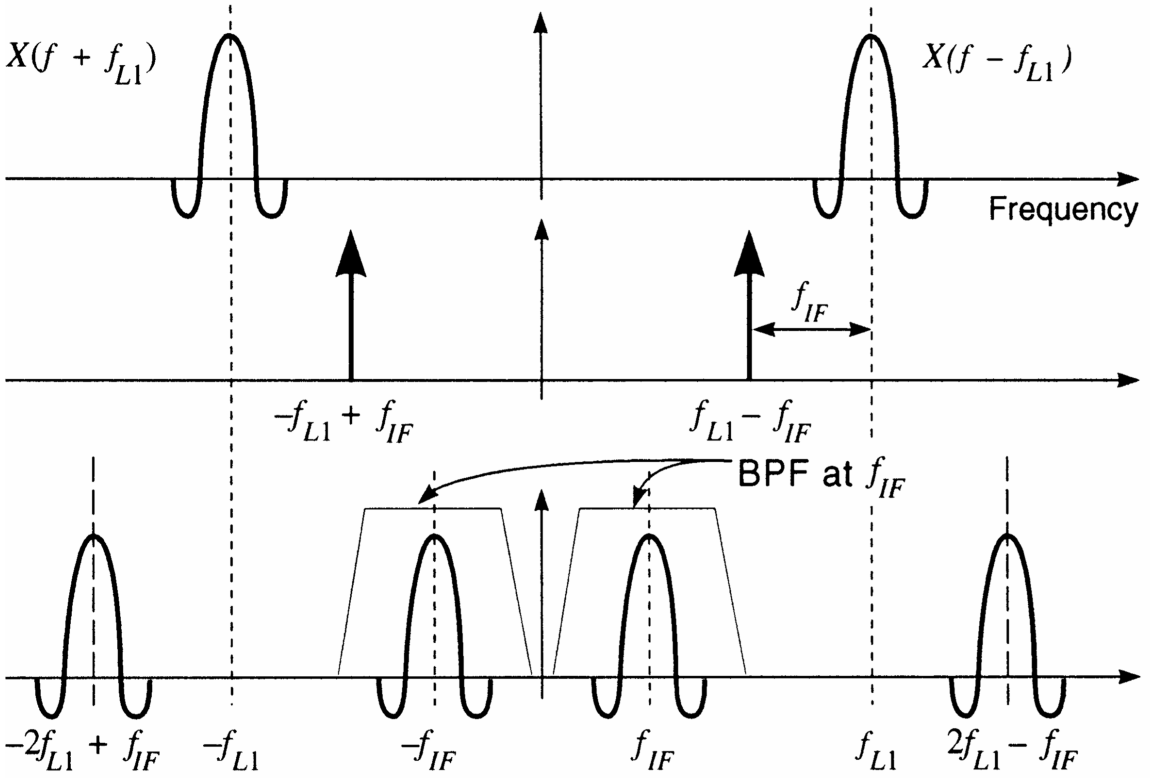


Figure 9.6 Fourier transforms before and after down conversion.

Power Spectrum and Auto-correlation

Power spectrum $$S(f) = |X(f)|^2$$

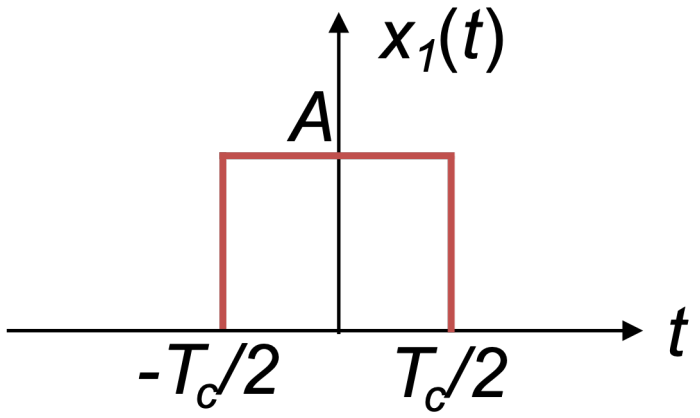
Auto-correlation $$R(\tau) = \int_{-\infty}^{\infty} x(t)x(t+\tau)dt$$

If $x(t)$ is real, $$S(f) = \int_{-\infty}^{\infty} R(\tau)e^{-j2\pi f\tau}d\tau$$

$$R(\tau) \overset{FT}{\Leftrightarrow} S(f)$$

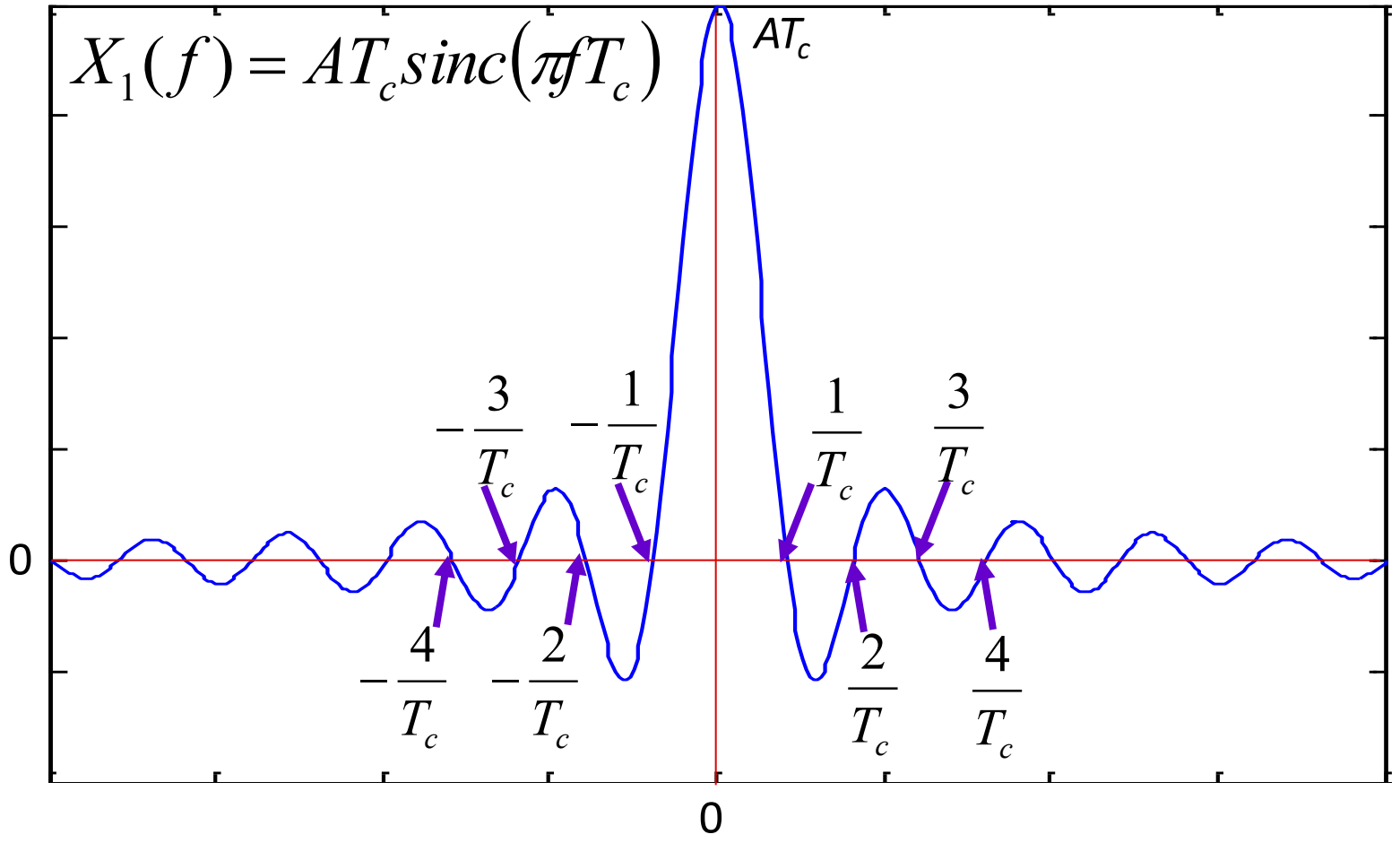
A signal's auto-correlation and power spectrum are a FT pair

Example 1: Rectangle Pulse Signal Spectrum



$$X_1(f) = AT_c \frac{\sin(\pi f T_c)}{\pi f T_c}$$

$$\mathrm{sinc}(\pi f T_c)$$



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Next lecture: GNS Spectrum